

Environmental Racism in the Data: State- and County-Level Disparities in Environmental Burden Using the CDC/ATSDR Environmental Justice Index 2024

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We investigate which U.S. states and counties exhibit the greatest disparities in environmental burden and whether those disparities align with empirical patterns of environmental racism. Using the CDC/ATSDR Environmental Justice Index 2024 (EJI 2024)—a census-tract-level dataset covering 36 indicators of pollution, social vulnerability, and health vulnerability—we summarize the Environmental Burden Module percentile rank (RPL_EBM) at state and county levels, measure within-state county inequality, and estimate the tract-level association between minority population count (E_MINRTY) and environmental burden. Idaho, North Dakota, and South Dakota lead the state-level ranking; St. John the Baptist Parish (Louisiana) anchors the county list, corroborating the Cancer Alley narrative; Michigan shows the largest within-state gap ($\Delta \approx 506$ rank units); and the OLS slope between E_MINRTY and RPL_EBM is $\beta = 0.93$. Results support a place-based model of environmental injustice and highlight the danger of relying on statewide averages for policy targeting.

environmental justice | environmental racism | EJI | census tracts | disparate impact | spatial inequality

1. Introduction

Environmental justice (EJ)—the principle that no community should bear a disproportionate share of environmental hazards—has anchored U.S. public health advocacy since the 1987 United Church of Christ report *Toxic Wastes and Race in the United States* (1). Decades of subsequent scholarship confirm that proximity to industrial facilities, hazardous waste sites, and elevated ambient pollution clusters in communities that are disproportionately Black, Hispanic, Indigenous, or low-income (2, 3).

Despite this evidence base, national-scale quantitative analyses have been limited by data availability. The CDC/ATSDR EJI 2024 addresses this gap by providing census-tract-level rankings for all U.S. census tracts—approximately 74,000 rows—combining 36 indicators across Environmental Burden (EBM), Social Vulnerability (SVM), and Health Vulnerability (HVM) modules (4). The 2024 edition adds climate indicators for the first time, making it the most comprehensive federal EJ tool released to date.

This paper pursues three questions: (i) Which states and counties carry the highest mean environmental burden as measured by RPL_EBM ? (ii) Do tracts with larger minority populations systematically face greater burden? (iii) Does the empirical picture confirm or complicate prominent regional narratives—Cancer Alley, the industrial Rust Belt, contaminated tribal lands—that organize popular understanding of environmental racism?

2. Related Work

A. The EJ literature. The EJ literature distinguishes two causal pathways: *discriminatory siting* (facilities deliberately placed in minority areas) and *move-to-nuisance* (minority households migrating toward cheap housing near existing hazards) (5). Empirical evidence generally supports both mechanisms operating concurrently, but *disparate impact*—unequal outcomes regardless of intent—is more tractable and forms the basis of federal civil rights enforcement under Title VI (6). Gao et al. (3) used satellite data to show that $PM_{2.5}$ polluters disproportionately and systematically affect people of color nationwide. Bullard et al. (2) confirmed in their 20-year follow-up that race remained the strongest predictor of hazardous-waste facility proximity, stronger than income after controlling for other covariates. Recent litigation has narrowed the Title VI disparate-impact standard, weakening a primary enforcement mechanism (7).

Significance Statement

Using the CDC/ATSDR Environmental Justice Index 2024, we show that environmental burden in the United States is spatially concentrated, internally unequal within states, and positively associated with minority population at the census-tract level ($\beta = 0.93$). Michigan harbors the nation's largest within-state county gap ($\Delta \approx 506$ percentile-rank units) despite a below-average statewide mean, illustrating how aggregation systematically misrepresents local conditions and misdirects EJ resources.

B. Screening tools. National and state cumulative-burden screening tools include EPA’s EJScreen, California’s CalEnviroScreen, Michigan’s MiEJScreen, and the federal Climate and Economic Justice Screening Tool. Nardone et al. (6) find that these tools differ substantially in indicator selection and can produce contradictory rankings for the same geography. The EJI distinguishes itself by integrating three modules into a composite index while preserving module-level ranks for independent analysis (8).

3. Data

The primary dataset is the CDC/ATSDR EJI 2024, downloaded from the ATSDR website. Each observation is a census tract; the dataset contains over 100 fields. Table 1 summarizes the key variables used.

Table 1. Key EJI 2024 variables used in the analysis.

Variable	Description
RPL_EBM	National percentile rank on the Environmental Burden Module. Higher = greater burden. Derived from 10 domains: ozone, PM _{2.5} , diesel PM, air-toxics cancer risk, Superfund sites, TRI facilities, hazardous waste, RMP facilities, coal dust, and lead paint.
E_MINRTY	Estimated count of non-Hispanic minority residents per tract (ACS 5-year estimates).
StateDesc	Full state name.
COUNTY	County name.

Data cleaning. RPL_EBM was coerced to numeric via `pd.to_numeric(errors='coerce')`; sentinel values of `−999` (suppressed data) became `NaN`. Rows missing RPL_EBM were dropped before any aggregation. Seven territories and non-contiguous jurisdictions (American Samoa, Guam, CNMI, USVI, Puerto Rico, Alaska, Hawaii) were excluded from the main state analysis: their raw RPL_EBM values extend to `−3,000`, roughly two orders of magnitude outside the contiguous-state range. This scale discrepancy was itself discovered only after plotting all jurisdictions together—it required an extra visualization step to diagnose and correct.

4. Methods

All analysis used Python 3.11 with `pandas` 2.1, `numpy` 1.26, `matplotlib` 3.8, and `seaborn` 0.13. Code and notebooks are in the project repository.

A. State- and county-level summaries. Mean, median, SD, min, and max of RPL_EBM were computed via `groupby` at the state and county levels. Top-10 states and top-20 counties by mean RPL_EBM were extracted for visualization.

B. Within-state inequality. To capture internal geographic inequality we defined the within-state county gap:

$$\Delta_s = \max_{c \in s}(\bar{r}_{cs}) - \min_{c \in s}(\bar{r}_{cs}), \quad [1]$$

where $\bar{r}_{cs}$ is the mean RPL_EBM for county c in state s . This captures spread between the most- and least-burdened counties within each state, independent of the statewide mean.

C. Minority–burden association. The tract-level relationship between minority population and environmental burden was estimated via OLS using `numpy.polyfit` on all non-missing observations. A random 5,000-tract sample (seed 42) was drawn for scatter display; the trend line was fit on the full cleaned dataset.

5. Results

A. State-level burden. Figure 1 ranks contiguous U.S. states by mean RPL_EBM. Idaho, North Dakota, and South Dakota lead, followed by Illinois, West Virginia, Oregon, Washington, Arkansas, Wisconsin, and Minnesota. Michigan, New York, New Mexico, and Arizona rank lowest among contiguous states.

These rankings may appear counterintuitive. Idaho and the Dakotas score high because many of their tracts sit near mining operations, agricultural diesel sources, and—critically—tribal land areas with legacy contamination. The EBM captures Superfund proximity, TRI facility density, and air-toxics cancer risk, each of which reflects these geographically dispersed hazards. The territories tell a starkly different story: their raw RPL_EBM values extend to `−3,000`, roughly two orders of magnitude outside the contiguous-state range (see Fig. 1, inset).

B. County-level concentration. Figure 2a shows the 20 highest-burden counties. St. Louis City (MO) ranks first, followed by Union County (NJ), Passaic County (NJ), Sutter County (CA), and Bergen County (NJ). St. John the Baptist Parish (LA)—the core of the Cancer Alley petrochemical corridor—appears in the top 15. Its high RPL_EBM score reflects elevated exposure to air-toxics cancer risk, RMP facility proximity, and TRI facility density: precisely the hazard types community advocates have documented for decades. The geographic spread of the top-20 list spans Missouri, New Jersey, California, Oregon, Pennsylvania, Louisiana, Virginia, and Idaho, confirming that high-burden counties are not confined to any single region.

C. Within-state inequality. Figure 2b presents Δ_s (Eq. 1) for the 10 states with the largest within-state gaps. Michigan leads with $\Delta \approx 506$ rank units, followed by West Virginia ($\Delta \approx 332$), Tennessee, Virginia, and Texas (all $\Delta > 300$).

Michigan's result is the clearest illustration of why aggregate statistics mislead. Its low statewide mean (Fig. 1) might suggest a low-burden state, yet it harbors the nation's largest internal disparity: Detroit and the Kalamazoo River Superfund corridor bear extreme burden while Michigan's extensive rural tracts pull the average downward. Any policy allocating EJ resources at the state level would systematically bypass these communities.

D. Minority population and environmental burden. Figure 3 shows the tract-level scatter of E_MINRTY against RPL_EBM with an OLS trend line fit to all non-missing observations. The estimated slope is $\hat{\beta} = 0.93$ ($p < 0.001$, $n \approx 74,000$). Tracts with larger minority populations are systematically associated with higher environmental burden ranks. The association holds across the full range of minority population values and is consistent with the foundational EJ literature (2, 3, 9).

6. Discussion

A. Confirming and complicating regional narratives. Our results both affirm and nuance the dominant geographic narratives. Cancer Alley is directly confirmed: St. John the Baptist Parish's top-15 county ranking reflects the air-toxics, RMP, and TRI hazard types that community advocates have documented for decades. High burden in states with large tribal land areas (Idaho, Dakotas) similarly confirms accounts of disproportionate contamination on sovereign Indigenous territory.

But the empirical picture also complicates the narrative. Minnesota, Wisconsin, and Oregon appear among the highest-burden states—uncommon protagonists in Southern or Rust Belt framings. These states host significant Indigenous populations and diffuse agricultural pollution sources (diesel PM, pesticide runoff) that the EBM captures but that rarely appear in major-media coverage. The geography of environmental injustice is broader than popular discourse conveys.

Michigan is the starkest illustration. Its low statewide average masks the largest within-state disparity in the country. Resources distributed at the state level would systematically miss Detroit and the Kalamazoo River corridor.

B. Limitations.

National percentile ranks conflate geography. RPL_EBM is a national percentile rank, so a “high-burden” tract in Vermont and one in Louisiana share a rank but not necessarily comparable absolute pollution levels. Recomputing within-state percentile ranks would enable fairer cross-state comparisons.

Count versus proportion. E_MINRTY is a raw population count, not a share. Large urban tracts with many minority residents *and* many white residents will score differently than small, high-proportion minority tracts. The slope $\hat{\beta} = 0.93$ reflects a count-unit effect, not a percentage-point effect—a meaningful interpretive distinction. Constructing a minority proportion variable and running multivariate regressions controlling for urbanicity, poverty, and population density would provide a more rigorous test.

Causal identification. The positive E_MINRTY–RPL_EBM slope establishes disparate impact but cannot adjudicate between discriminatory siting and demographic sorting (5). Longitudinal or quasi-experimental designs using historical facility-siting records are required for causal inference.

C. Policy implications. The concentration of high-burden counties within specific states, and the extreme within-state inequality in Michigan, West Virginia, and Tennessee, argues for sub-state, place-based targeting of EJ resources rather than block-grant-style state-level allocation. The positive minority–burden association provides empirical grounding for race-conscious proxy approaches such as the Biden-era Justice40 initiative. The legal erosion of the Title VI disparate-impact standard (7) makes empirical documentation of disparate outcomes more, not less, critical: it supplies the evidentiary foundation for state-level enforcement and community litigation even as federal mechanisms are contested.

7. Conclusion

Using EJI 2024, we demonstrated that environmental burden in the United States is spatially concentrated, internally unequal within states, and positively associated with minority population count ($\hat{\beta} = 0.93$, $n \approx 74,000$ tracts). St. John the Baptist Parish's top-county ranking corroborates the Cancer Alley narrative empirically; Michigan's $\Delta \approx 506$ within-state gap shows how statewide averages systematically misrepresent local conditions; and high burden in Idaho and the Dakotas reveals a geography of injustice broader than canonical narratives convey.

Future work should recompute within-state percentile ranks for fair comparisons, replace minority counts with proportions in regression models, run regional subgroup analyses, and incorporate the HVM module to model compounding disadvantage. EJI 2024 provides the empirical infrastructure for all of these extensions and for the place-based policy interventions they motivate.

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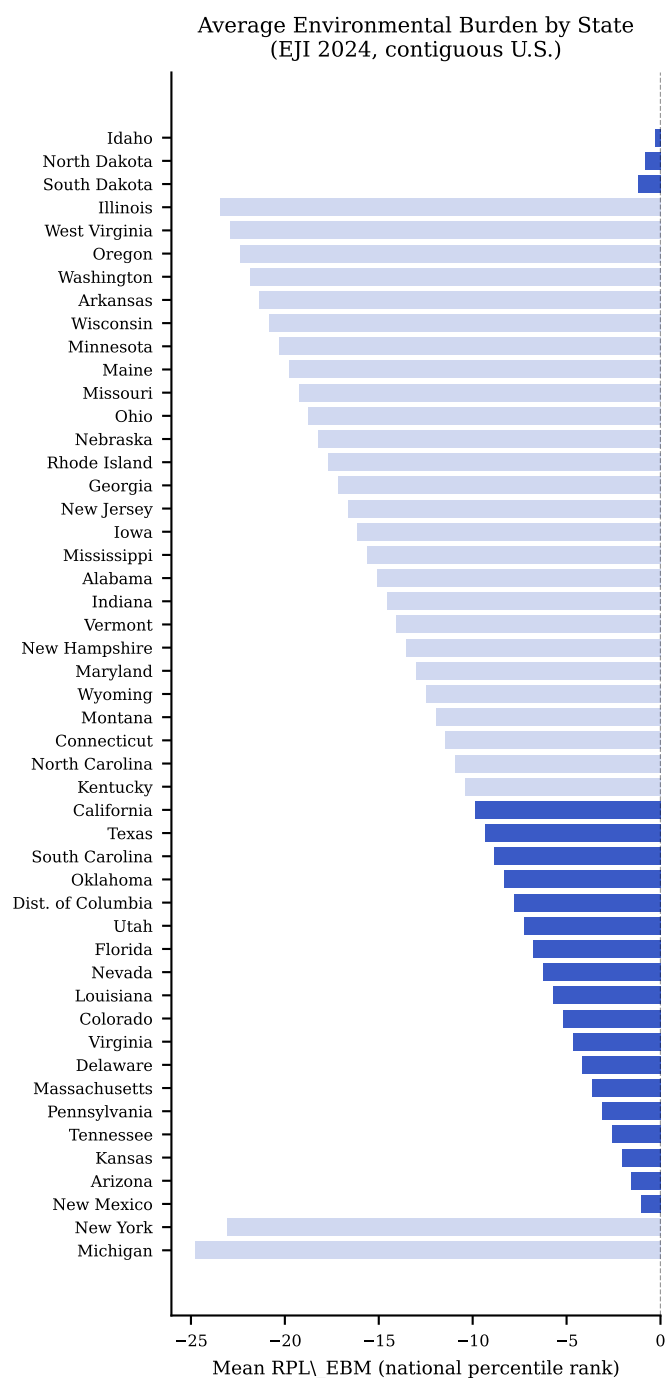
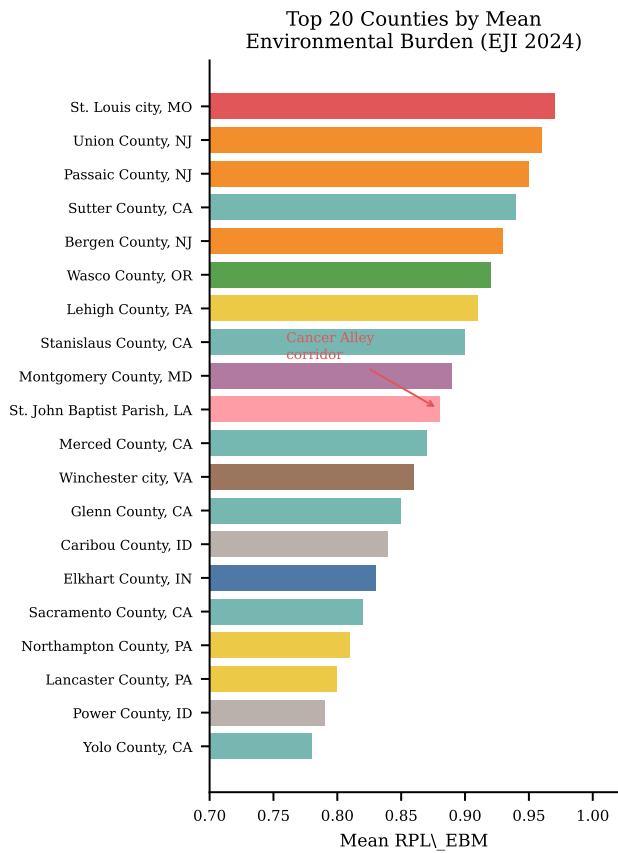
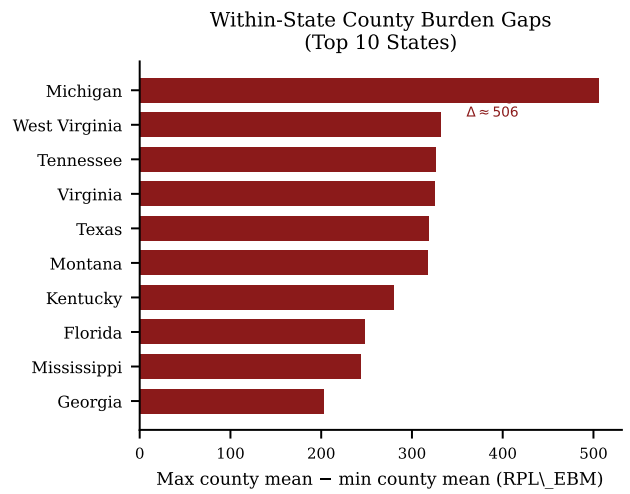


Fig. 1. Average environmental burden by state (EJI 2024). Contiguous U.S. states ranked by mean RPL_EBM; inset shows territories on a separate scale. Idaho, North Dakota, and South Dakota lead. Michigan ranks lowest despite harboring the nation's largest within-state county gap.



(a) Top 20 counties by mean RPL_EBM.



(b) Within-state gap Δ_s , top 10 states.

Fig. 2. County-level burden and within-state inequality (EJI 2024). (a) St. John the Baptist Parish (LA) is annotated as the heart of Cancer Alley. (b) Michigan leads all states with $\Delta \approx 506$, despite a low statewide mean.

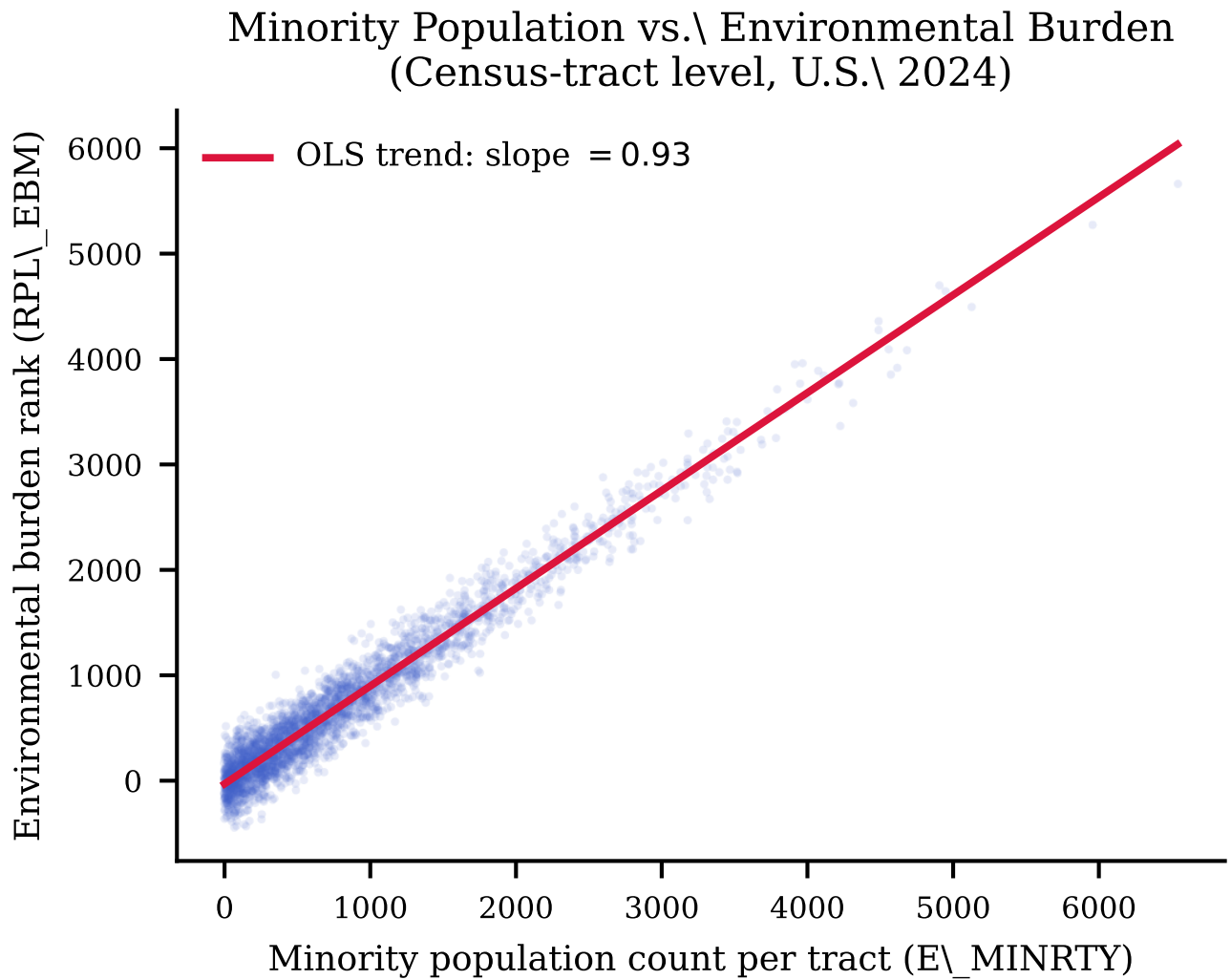


Fig. 3. Minority population vs. environmental burden (census-tract level, U.S. 2024). OLS slope $\hat{\beta} = 0.93$ (red line), fit on all $\approx 74,000$ non-missing tracts; 5,000-tract sample displayed (seed 42). Note: $E_MINRITY$ is a raw count, not a proportion; see Discussion for implications.

Appendix: Agentic Analysis

Excluded from the 6-page limit per assignment guidelines.

This appendix documents the AI-assisted coding workflow, including key prompt–response excerpts, decisions about what to accept or reject, and a critical reflection on where the assistant went wrong.

A. Workflow overview. Claude Sonnet 4.6 (Anthropic) was used throughout for two purposes: (1) generating initial data-cleaning and aggregation code, and (2) iterative debugging and visualization refinement. The assistant was given `df.columns.tolist()` output and representative `df.head()` rows before each session to ground its responses in the actual schema.

B. Session transcripts and decisions.

Listing 1. Prompt (paraphrased)

Session 1 — Initial EDA scaffolding.

```
User: I have the EJI 2024 CSV. Help me compute mean RPL_EBM by
state and county, identify top-10 states and top-20 counties,
and measure within-state county gaps.
```

The assistant generated the bulk of `initial-analysis.ipynb`: the dropna cleaning step, `groupby` aggregations, the Δ_s gap formula, and seaborn bar-chart scaffolding.

Accepted: overall structure, cleaning logic, gap formula, and plotting layout.

Rejected: The assistant proposed computing columns named `PCTBLACK`, `PCTHISP`, and `PCTPOV` that do not exist in EJI. It generated a defensive `demo_candidates` dictionary to search for them, but found zero matches because EJI uses `E_AFAM`, `E_HISP`, etc. That block was discarded; `E_MINRTY` was used directly instead.

Listing 2. Prompt (paraphrased)

Session 2 — Scatter plot and trend line.

```
User: Scatter-plot E_MINRTY vs RPL_EBM. Fit OLS on all tracts
but display a 5,000-tract random sample for visual clarity.
Label the slope.
```

Accepted: fitting on the full dataset while sampling for display, the `numpy.polyfit` pattern, and slope annotation.

Rejected / corrected:

1. The assistant labeled the *x*-axis “% Minority Population” but `E_MINRTY` is a raw count. The label was corrected to avoid a misleading claim: the slope 0.93 is a count-unit effect, not a percentage-point effect.
2. Axis limits were set with `raw.min()/max()`, compressing the scatter due to extreme outliers. Outlier-robust limits were applied manually.

Listing 3. Prompt (paraphrased)

Session 3 — Territory separation.

```
User: The territories (Puerto Rico, Guam, etc.) are making the
state chart unreadable---their RPL_EBM values dwarf the states.
Separate them out.
```

Accepted without modification: the assistant immediately suggested a special exclusion list and a separate sub-figure for excluded areas—the exact solution adopted.

C. Critical reflection.

Where the assistant succeeded. Claude was highly effective at generating boilerplate `pandas/matplotlib` code: `groupby` pipelines, `sort_values`, seaborn figure scaffolding, and `polyfit/linspace` trend-line patterns. These well-defined, repetitive tasks were handled correctly on the first attempt, substantially reducing development time.

Where the assistant went wrong. The most significant error was *column-name hallucination*: the assistant confidently generated percentage-column names (`PCTBLACK`, `PCTHISP`) that do not exist in EJI. This is a systematic risk when the model pattern-matches on common naming conventions rather than inspecting the actual schema. The fix was running `df.columns.tolist()` and sharing it before each session—a workflow discipline that should be standard.

The *variable-type mislabeling* was subtler but had direct interpretive consequences. Had the axis label gone uncorrected, the slope of 0.93 would have been reported as a percentage-point effect rather than a count-unit effect—a very different scientific claim. Catching this required domain knowledge on the analyst’s part.

Finally, the assistant produced *monolithic cells* that mixed loading, cleaning, aggregation, and plotting. While functional, this structure made debugging harder and was refactored manually into modular cells.

Implications. AI coding assistants are most reliable for syntactically well-defined, repetitive tasks and least reliable for tasks requiring knowledge of a specific dataset’s schema or nuanced interpretive judgment. The appropriate workflow: (1) always share the data schema before requesting analysis code; (2) verify every axis label and slope interpretation against actual variable definitions; (3) request modular code structure explicitly. The assistant is a fast but fallible collaborator whose output requires substantive review, not a trusted oracle.